

# LINFENG JIANG

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## EDUCATION

### University of Maryland, College Park

Expected May 2027

B.S. in Information Science + Advanced Cybersecurity | iSchool | ACES Honors College (Advanced Cybersecurity Experience for Students)

GPA: 3.96 / 4.00 | Semester Academic Honors, Fall 2024-Spring 2026

**AI-focused coursework:** Generative AI, Data Science Techniques, Statistics for Information Science, Data Sources & Manipulation, Data Visualization, Object-Oriented Programming

## PUBLICATIONS, INTELLECTUAL PROPERTY & RESEARCH WRITING

- **"Let the Carrier Carry the Attack: Preserving the Subject in Adversarial Image Generation."**  
First author; ICLR 2027 submission (under review); arXiv:2609.39723.
- **"Rethinking Inter-LoRA Orthogonality in Adapter Merging: Insights from Orthogonal Monte Carlo Dropout."**  
AISTATS 2027 submission (under review).
- "Overview of PDF Security Principles," CONF-CDS, 2023.
- Heterogeneous Multi-Core Task Scheduling Method and System Based on an Improved Whale Optimization Algorithm, patent/application reference MC23-P10075.

## RESEARCH EXPERIENCE

### Concept-Based Adversarial Generation (Advisor: Andi Zhang)

Feb 2026-Present

- Developed a segmentation-guided framework for personalized concept-based adversarial generation, preserving subject identity while enabling controllable manipulation of editable background regions.
- Introduced the concept of semantic carriers to characterize visible target-related evidence in adversarial generation, and designed controlled carrier conditions to study their effects on attack success, transferability, and subject preservation.
- Designed an inversion-based adversarial guidance method that injects targeted classifier gradients along diffusion reconstruction trajectories for controllable concept-level attacks.

### SNN Robustness & Reprogramming (Advisor: Suyu FNU)

Feb 2026-Present

- Investigating whether temporal dynamics and spike-based computation provide natural adversarial robustness across SNN training/conversion methods, datasets, model architectures, and attack families; MNIST is the current experimental tested.
- Implemented matched ANN/SNN evaluation code and groupwise NRPM modules that reprogram synaptic contributions using representation and temporal discrepancy signals.
- Developing detect-then-reprogram and conditional-defense variants while studying robustness, attack success, detector behavior, latency, spike activity, and operation/energy cost.

### NIST AI Risk Management Accessibility | Research Project

Jan 2026- March

- Exploring concise inductive and deductive approaches for making NIST AI risk-management concepts easier to apply in practical model assessment.

## TEACHING EXPERIENCE

### Undergraduate Teaching Assistant, INST326: Object-Oriented Programming II | University of Maryland

Fall 2025-Present

### Undergraduate Teaching Assistant, INST126: Object-Oriented Programming I | University of Maryland

Fall 2025-Present

**Teaching focus:** Python, object-oriented programming, data structures, software design, problem solving, and assignment support.

## SELECTED TECHNICAL PROJECTS

### Ultra-Short-Term Photovoltaic Output Forecasting (CNN- LSTM)

Aug- Nov 2025

### Context-Aware Steganographic Text Generation

Sep- Dec 2025

### Centralized Security Dashboard | HACS408P Project

Spring 2026

## HONORS

Semester Academic Honors / Dean's List at UMD (2024-2026); top placements in JUST CTF, picoCTF, and Right CTF (2022); USACO Platinum (2022)

## TECHNICAL SKILLS

**Programming & data:** Python, C++, Java, R, SQL, MATLAB, Linux/UNIX, data analysis and visualization, PyTorch, etc.